Machine Learning Foundations: Geometry, Optimization, and Linear Classification

From vectors and planes to loss functions, feature vectors, and classifiers

Dr. Yael Demedetskaya

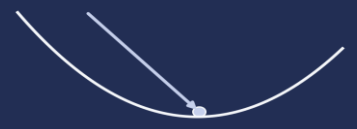
Point / Vector

$x=(3,2,2)$

Plane / Separator

$3x_1+x_2-1=0$

Loss / Optimization



$\theta \leftarrow \theta-\gamma\nabla L(\theta)$

Classifier

$h(x)\in\{-1,+1\}$

Geometry gives us representation

Optimization gives us learning

Classification gives us prediction

Foundations for early machine learning concepts and linear classification

Part I — Framing, Notation, and Roadmap

How the course’s mathematical language becomes machine learning language

vector — hyperplane — classifier

Dr. Yael Demedetskaya

Why These Foundations Come First

Linear Algebra / Geometry

Calculus / Optimization

Machine Learning

Probability / Uncertainty

Computation / NumPy

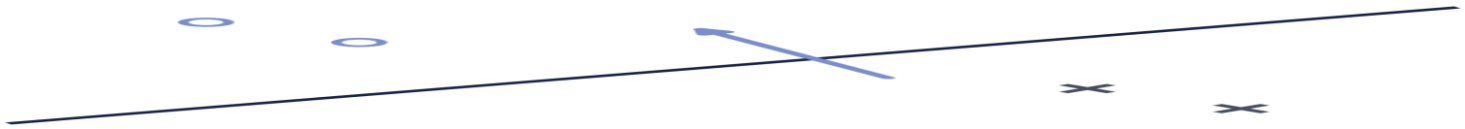
Core idea

Machine learning depends on representation, error measurement, optimization, uncertainty, and computation.

Representation, error, improvement, uncertainty, and computation all arrive before learning can happen.

Geometry Gives Us Representation

Geometry lets us place examples in space, compare directions, and define boundaries



vectors represent objects · space gives structure · boundaries separate regions

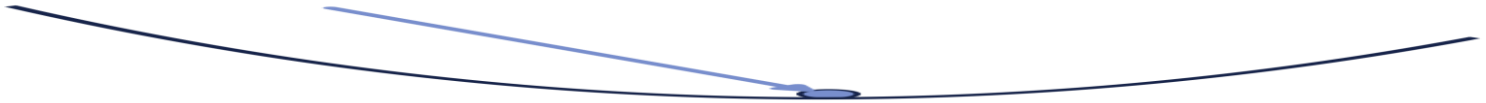
$x \in R^d \quad \cdot \quad \theta \cdot x + \theta_0 = 0$

point/vector $x=(3,2,2)$ and boundary $3x_1+x_2-1=0$

Geometry is the language that lets examples become locations, directions, and separable regions.

Optimization Improves Parameters

Optimization gives us a rule for improving parameters by moving toward lower loss



error becomes a function · lower is better · updates move downhill

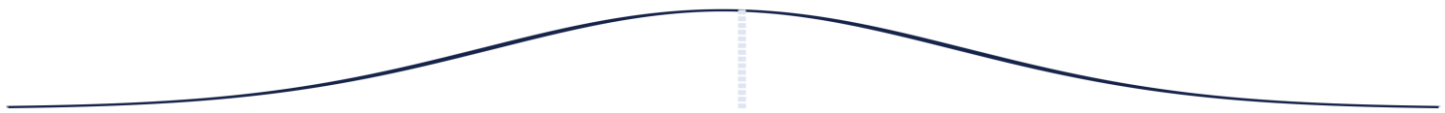
$L(\theta) \quad \cdot \quad \theta \leftarrow \theta - \gamma \nabla L(\theta)$

current parameter, lower-loss direction, visible minimum

Optimization turns model improvement into a concrete downhill search problem.

Probability Models Uncertainty

Probability lets us describe uncertainty, variability, and noisy data



outcomes can vary · uncertainty can be modeled · ML must reason under uncertainty

$$X \sim N(\mu, \sigma^2)$$

visible mean and spread

Probability gives machine learning a principled way to reason about noise and uncertainty.

Computation Makes the Math Executable

Computation turns the math into something we can evaluate, repeat, and optimize

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

vectors become arrays · formulas become code · repeated updates become algorithms

$$A\,x$$

matrix/vector/array structure as executable math

Computation is what lets mathematical ideas become trainable algorithms.

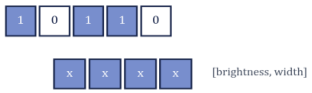
The Big Story: From Data to Decisions

Machine learning turns real-world objects into numerical representations, places them in a geometric space, and learns rules that predict labels or values.

Raw Object

raw input

Feature Vector

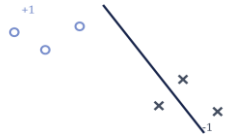


$x \in R^d$

Point in Space

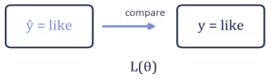
point / vector

Model / Boundary



$\theta \cdot x + \theta_0$

Prediction / Error



$h(x), L(\theta)$

Representation comes first: objects become vectors, vectors enter geometry, and geometry supports prediction.

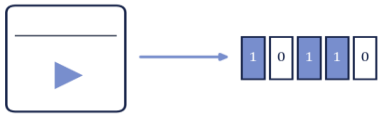
Example A — From Movie to Feature Vector to Label

A real object becomes a machine-readable representation, then a prediction.

Movie

object / metadata

Encoding



5-feature binary vector

$x=[1,0,1,1,0]$

Geometric Representation

point in feature space

Prediction

like / dislike

$h(x)=\text{sign}(\theta \cdot x+\theta_0)$

object • encoding • representation • prediction

A real object becomes a feature vector before it becomes a prediction.

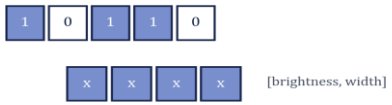
Example B — From Digit Image to Class Label

Different objects can still enter the same machine-learning pipeline once represented numerically.

Digit

image input

Features



brightness / width

x=[brightness,width]

Geometric Representation

point in low-dimensional space

Prediction

class label

h(x)=7

image • features • point • class

Different data types still pass through the same representation-first pipeline.

Running Examples We Will Reuse

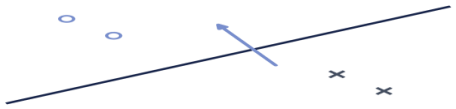
These examples will return throughout the deck as anchors.

Point / Vector

$x=(3,2,2)$

geometry anchor

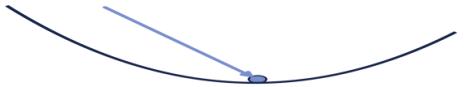
Plane / Hyperplane



$3x_1+x_2-1=0$

boundary anchor

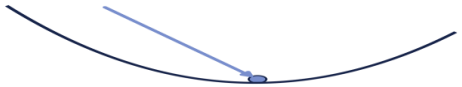
1D Optimization



$f(x)=\frac{1}{3}x^3-x^2-3x+10$

optimization anchor

Gradient



$f(\theta)=\theta_1^2+\theta_2^2$

descent anchor

Gaussian

$X\sim N(1,2)$

probability anchor

Movie Recommender

5-feature vector + label ±1

supervised-learning anchor

geometry, optimization, probability, computation, supervised learning

Notation You Must Read Fluently — Objects and Coordinates

In machine learning, the same letters are reused across different mathematical objects. Context matters.

symbol	meaning	visual example
$\mathbf{x}$	point / feature vector / example	$x=(3,2,2)$
$\mathbf{x_i}$	i-th coordinate	$x_2=2$
$\mathbf{A}$	matrix	A x
$\mathbf{y}$	label	$y=+1$

$x \in R^d$ • $A \in R^{m \times n}$ • $y \in \{-1,+1\}$

Notation You Must Read Fluently — Parameters, Classifiers, and Loss

These symbols will appear repeatedly in linear models, classification, and optimization.

symbol	meaning	visual example
θ	parameter vector / normal vector	$\theta \cdot x + \theta_0 = 0$
θ_0	offset / intercept	boundary shift
h	classifier / prediction rule	$h(x) = \mathrm{sign}(\theta \cdot x + \theta_0)$
L	loss	$L(\theta)$

$\theta \in R^d$ • $\theta \cdot x + \theta_0 = 0$ • $h(x) = \mathrm{sign}(\theta \cdot x + \theta_0)$ • $L(\theta)$

Point, Vector, Feature Vector: Same Shape, Different Role

Point

location in coordinate space

$x=(3,2,2)$

Vector

arrow from the origin

$x \in R^3$

Feature Vector

encoded example for prediction

$x \mapsto$ feature vector

location in space • displacement from origin • encoded example for prediction

$x=(3,2,2)$ • $x \in R^3$ • $x \mapsto$ feature vector

One Object, Three Interpretations

The coordinates did not change. The interpretation changed.

x=(3,2,2)

Point

same coordinates, location in space

same coordinates

Vector

same coordinates, displacement from origin

different role

Feature Vector

same shape, encoded movie example

context changes meaning

x=(3,2,2) • movie → x

The shape of the mathematical object stays the same; only the interpretation changes.